Budgeted Page-Level Preprocessing Selection for Vietnamese Administrative OCR: A Controlled Comparison of Learned, Retrieval-Based, and Static Shortlists

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Abstract

Purpose: We study page-level preprocessing selection for administrative-document OCR under a limited OCR-call budget. **Methods:** A stratified five-parameter design produces 50 configurations. Recognizer-specific CER regressors rank them using OCR-free page features and configuration parameters. Nested GroupKFold compares surrogate, nearest-neighbor transfer (NN-K), FixedTopK, and random shortlists on 300 Vietnamese administrative first pages, with and without contour suppression, and 612 SROIE receipts. An oracle-free selector chooses one output from the surrogate top-3. **Results:** After Holm correction within each policy-contrast/evaluation-block family, Surrogate and NN-K outperform random shortlists across all recognizers, settings, and budgets, and the surrogate outperforms FixedTopK in all eight SROIE best-of-K comparisons. FixedTopK remains competitive on Vietnamese data. Headroom correlates with the other three policies' FixedTopK-relative gains; Random's correlation exceeds Surrogate's in 30/32 cases, supporting a policy-agnostic interpretation. The strict selector yields lower mean CER than the static single-best configuration (FixedBest) in five of eight settings, with three significant differences after Holm correction. **Conclusion:** No shortlist policy is uniformly best. Page-conditioned shortlists outperform random sampling; the selector requires no reference text at inference time.

Keywords: document OCR, preprocessing selection, algorithm selection, surrogate-based ranking, nearest-neighbor transfer, oracle-free output selection

1 Introduction

Optical character recognition (OCR) on Vietnamese administrative documents is sensitive to image preparation. Stamps, headers, and logos can generate poor text crops, while the effectiveness of denoising, contrast enhancement, brightness correction, and sharpening varies across pages.

Exhaustive per-page preprocessing search requires an OCR pass for every candidate. We instead ask whether a model trained on historical pages can identify a promising shortlist for a new page under a limited OCR budget.

Using initial PaddleOCR detection and three OCR backends, we compare learned, retrieval-based, and static shortlists from a fixed pool of 50 preprocessing candidates. OCR-free page features condition the learned and retrieval-based policies; observed candidate CER supplies their offline training and evaluation targets.

Nested, page-grouped comparisons evaluate shortlist quality under equal OCR-call budgets and test whether a top-3 selector can choose one output without reference text at inference. Random and NN-K controls distinguish policy-specific evidence from the shared-baseline correlation between FixedTopK-to-oracle headroom and FixedTopK-relative gains.

The paper makes the following contributions:

1. It formulates page-level OCR preprocessing as a budgeted shortlist-selection problem over a fixed candidate pool and evaluates it across recognizers, preprocessing conditions, and a second document domain.
2. It provides a controlled nested GroupKFold comparison of cross-instance surrogate ranking, nearest-neighbor transfer, static FixedTopK, and random shortlist policies under identical OCR-call budgets.
3. It evaluates a nested oracle-free top-3 output selector that selects one OCR result from a surrogate shortlist without reference text at inference time.
4. It provides controlled diagnostics of FixedTopK-to-sampled-oracle headroom using shared-baseline and direct policy-to-policy controls, separating retrospective improvement potential from policy-specific evidence.

Our earlier conference paper [2] combined PaddleOCR, VietOCR, stamp handling, and language-model post-processing in an end-to-end pipeline. This study instead investigates budgeted preprocessing selection, adding a fixed candidate pool, surrogate and retrieval baselines, nested evaluation, oracle-free output selection, and a second document domain (SROIE).

2 Related Work

2.1 OCR Systems and Document Preprocessing

Modern modular OCR pipelines, including PaddleOCR/PP-OCR, separate detection from recognition, with detected text regions cropped before recognition [4, 10]. Our pipeline follows this structure with PaddleOCR detection and three alternative recognizers based on different design assumptions: Tesseract [24], EasyOCR [8], and the Vietnamese-specific VietOCR [17]. Vietnamese document analysis and OCR is an active application area facing complex diacritics, limited annotated data, and real-world document variation [9], with representative work spanning scene-text recognition and the VinText benchmark [13], administrative-document digitization [14], and MC-OCR receipt recognition [31]. Against this background, this paper studies adaptive page-level preprocessing selection as an orthogonal axis to recognizer choice; the recognizers themselves remain unmodified.

Document image preprocessing has a long history in OCR. Global thresholding such as Otsu's method [15] and local/adaptive methods such as Sauvola and Pietikäinen [22] are canonical baselines, with DIBCO-style evaluations established as benchmarks for historical and degraded documents [18]. Beyond binarization, OCR pipelines often use photometric enhancement and denoising operations such as contrast-limited adaptive histogram equalization (CLAHE) [32], bilateral filtering [28], gamma correction, and unsharp masking; these directly motivate the parameter space evaluated here. Recent learned restoration methods, including DE-GAN [26] and DocEnTr [27], learn image-to-image enhancement transforms and are complementary to our setting, which instead learns a cross-instance surrogate that ranks lightweight preprocessing parameter settings by downstream OCR CER. Administrative documents also contain artifacts such as stamps, decorative headers, and watermarks; this motivates the contour-suppression setting evaluated here, while the main contribution remains budgeted prioritization of preprocessing candidates rather than artifact removal.

2.2 Algorithm Selection, Meta-Learning, and Surrogate Optimization

Per-instance algorithm selection uses instance features to choose among alternatives with instance-dependent performance [20]. Our fixed-pool shortlist formulation is related, but selects multiple preprocessing candidates under an OCR budget rather than a single algorithm. Surrogate and sequential model-based optimization methods are also relevant to our setting. Methods such as SMAC [7], Bayesian optimization frameworks [25], and TPE-based sequential optimization [1] guide a sequential optimization run: configurations are evaluated iteratively, and the search model or distribution is updated from observations collected during that run. In standard formulations, this process is optimized for the current task rather than reused unchanged to rank a fixed pool on unseen document pages. In contrast, our surrogate is trained once across many document pages and reused at inference to rank preprocessing candidates for a new page without per-image OCR feedback.

Our data collection strategy is also closer to design-of-experiments than to iterative Bayesian optimization. Space-filling designs such as Latin hypercube sampling [12] are useful for broadly covering the preprocessing parameter space during offline data collection.

Closer to adaptive preprocessing, Garg and Chaudhury [5] studied automatic selection of document-image enhancement parameters using image-quality assessment and expected recognition accuracy within an EM-based optimization framework. Vats et al. [30] applied Bayesian optimization to hyperparameter selection in document image binarization, and Singh et al. [23] learned surrogate models for document image quality metrics to support automated document image processing. Rabeux et al. [19] predicted binarization results from document-degradation features to select a binarization method for each image; we instead rank a fixed pool of photometric settings by predicted OCR CER and evaluate shortlists under an OCR-call budget. These studies support the relevance of adaptive parameter selection and surrogate modeling in document preprocessing. However, the studies by Garg and Chaudhury and by Vats et al. focus on parameter optimization and binarization, whereas Singh et al. use cross-image surrogate modeling for document-image quality metric estimation rather than OCR-CER-based shortlist ranking.

2.3 Positioning of This Work

Our contribution is budgeted shortlist selection, not a new OCR engine, enhancement transform, or iterative per-page optimizer. The fixed-pool formulation compares static, retrieval-based, and learned policies by their ability to prioritize useful candidates rather than predict CER exactly.

The results show informative page-conditioned shortlists but no universally best policy: FixedTopK remains competitive, headroom is policy-agnostic, and the nested selector recovers only part of the shortlist gain. To our knowledge, the joint formulation of budgeted OCR-preprocessing shortlists, controlled learned/retrieval/static/random comparisons, and nested oracle-free output selection has not been explicitly studied in prior OCR-preprocessing work.

3 Dataset and Task Definition

3.1 Vietnamese Administrative Document Dataset

The primary dataset consists of 300 Vietnamese administrative document images obtained from the Vietnamese Government portal (https://chinhphu.vn/he-thong-van-ban) and spanning the years 2015 to 2024. At collection time, the first 30 documents appearing in the portal listing for each year were selected. An exact-content audit of the initially selected cohort identified two duplicated selections; these were replaced by one additional document from the corresponding 2018 and 2024 collections before the reported experiments were recomputed, preserving 30 images per year. The final image identifiers and image/reference SHA-256 hashes are recorded in the public cohort manifest. Source documents were treated at the page level: when a source PDF contained multiple pages, only the first page was converted to an image and used in this study. Thus, each sample corresponds to a first-page document image rather than a complete multi-page document. Reference transcriptions were manually prepared

from all 300 page images by a team member who did not consult OCR outputs. Targeted quality checks and corrections were subsequently performed before the analyses; these supplementary checks are distinct from the initial manual transcription. Receiving stamps added by the publishing portal were not transcribed. Before evaluation, OCR outputs and reference transcriptions were both normalized to Unicode NFC, and all whitespace sequences, including tabs and line breaks, were collapsed to single spaces. No case folding or punctuation removal was applied to the Vietnamese data. Because the corpus is drawn from a single public government document portal, it captures temporal variation in document appearance and image quality while limiting broader source-domain variation.

3.2 Task Definition

The task is defined as follows: given a document page image, produce OCR text for the full page. Performance is evaluated by page-level character error rate (CER) against the corresponding reference transcription. The optimization target is end-to-end page-level CER: preprocessing parameters are applied to the full page before text detection, so the objective captures effects on both detection quality and recognition accuracy, rather than recognition-only performance on pre-cropped text regions.

3.3 OCR Pipeline

The OCR pipeline used throughout the study consists of optional contour suppression on the full page, page-level preprocessing, initial PaddleOCR text-region/text-line detection, processing of the resulting crops by one of the three OCR backends, and page-level CER computation against the reference transcription. PaddleOCR provides the initial detection stage shared across backends, with its settings fixed within each dataset. EasyOCR additionally detects text inside each crop through `readtext`, and Tesseract performs internal page segmentation; this is therefore not a recognizer-only comparison with identical detection throughout. Preprocessing is applied before the initial detection stage, so it can affect both the text regions found by PaddleOCR and subsequent crop processing. Page-level artifacts such as stamps, decorative headers, logos, and watermarks may cause false, missed, or low-quality detections; once text is missed or incorrectly cropped, a downstream recognizer cannot fully recover it. The study therefore optimizes preprocessing at the page level rather than as a recognition-only crop enhancement step.

3.4 SROIE Receipt Dataset

The second dataset is the ICDAR 2019 SROIE benchmark [6], used in a different domain and OCR language configuration. Filename-level filtering excludes names containing `(` and requires matching image/transcription stems, yielding 626 receipts. A SHA-256 audit found five byte-identical image pairs with inconsistent benchmark transcriptions; both members of each ambiguous pair were excluded, leaving 616. A later image-identity screen and targeted visual review identified two further same-receipt pairs with different image bytes. Both members of each were excluded, without selecting a representative by OCR performance, leaving 612 receipts. All affected SROIE training layers and comparisons were recomputed on new page-group folds; old/new differences therefore include changed partitions, not only duplicate removal. Exclusion IDs, hashes and screening scope are released; the screen does not guarantee exhaustive duplicate recall. Bounding-box annotations are not used for CER. English configurations are PaddleOCR `lang=en`, Tesseract `eng`, and EasyOCR `en`; Vietnamese-specific VietOCR is omitted. This evaluates full-page shortlist ranking, not SROIE field extraction. For SROIE, the primary metric was changed from case-sensitive to casefolded CER after initial results and a best-of-K sensitivity analysis had been obtained, following inspection of annotation letter-case fidelity. Online Resource 1 retains the chronology and a matched case-sensitive sensitivity on the corrected cohort; earlier outputs remain archived. Case-insensitive SROIE edit-distance evaluation also has precedent in Rotman et al. [21], whose scoring procedure is not otherwise identical to ours.

4 Proposed Method

Fig. 1 summarizes the proposed framework, separating offline candidate evaluation and policy construction from inference-time shortlist construction, best-of-K shortlist evaluation, and oracle-free top-3 output selection.

4.1 Problem Formulation

Let $\mathcal{D} = \{(x_i, r_i)\}_{i=1}^{N}$ denote the page images and their reference transcriptions. The primary Vietnamese dataset has $N = 300$, and the SROIE receipt dataset has $N = 612$ after filename-level, exact-content and same-receipt filtering. Let $\Theta = \{\theta_1, \ldots, \theta_M\}$ denote the fixed candidate pool of page-level preprocessing settings, with $M = 50$. For a given experimental setting, define the page representation as $\tilde{x}_i = x_i$ in the raw setting and $\tilde{x}_i = C(x_i)$ in the contour-suppressed setting, where C is the fixed contour-suppression transform described below.

For recognizer e, the observed OCR response for page i and preprocessing candidate θ_j is

$$y_{i,j,e} = \mathrm{CER}\left(O_e(P_{\theta_j}(\tilde{x}_i)), r_i\right), \tag{1}$$

where P_{θ_j} applies the page-level preprocessing candidate and O_e denotes the OCR pipeline with initial shared detection and OCR backend e. Page-level CER is computed from edit distance:

$$\mathrm{CER}(o, r) = \frac{S(o, r) + D(o, r) + I(o, r)}{|r|}, \tag{2}$$

where S, D, and I are the numbers of substitutions, deletions, and insertions needed to transform OCR output o into reference transcription r.

For each page i and preprocessing candidate θ_j, the surrogate input is

$$z_{i,j} = [f(\tilde{x}_i), \theta_j, g(f(\tilde{x}_i), \theta_j)] \in \mathbb{R}^{22}, \tag{3}$$

where $f(\tilde{x}_i)$ contains 12 OCR-free page features, θ_j contains the 5 preprocessing parameters, and $g(f(\tilde{x}_i), \theta_j)$ contains 5 predefined interaction terms between page properties and preprocessing controls, all defined in the remainder of Section 4; the 22-dimensional input thus comprises these three groups concatenated. A recognizer-specific surrogate h_e is trained as a supervised CER regressor that predicts $\hat{y}_{i,j,e} = h_e(z_{i,j})$. Although trained as a CER regressor, the surrogate is used at inference time as a ranking function. The induced page-conditioned top-K shortlist policy is

$$\pi_K^{h_e}(\tilde{x}_i) = \operatorname*{Kmin}_{\theta_j \in \Theta} h_e(z_{i,j}), \tag{4}$$

where Kmin returns the K preprocessing candidates with the lowest predicted CER values.

The practical objective is therefore not exact CER prediction, but page-conditioned shortlist construction under a fixed OCR budget:

$$\min_{\pi_K} \mathbb{E}_i \left[\min_{\theta_j \in \pi_K(\tilde{x}_i)} y_{i,j,e} \right], \quad \pi_K(\tilde{x}_i) \subset \Theta, \quad |\pi_K(\tilde{x}_i)| = K. \tag{5}$$

This is an induced selection objective rather than the direct training loss. The surrogate is fitted to observed CER values, and its ranking is evaluated by the best observed CER within the selected top-K shortlist.

The reference baselines can be expressed in the same finite candidate pool. The sampled-pool oracle is

$$B_{i,e}^{\mathrm{oracle}} = \min_{\theta_j \in \Theta} y_{i,j,e}, \tag{6}$$

which is the lowest observed CER within the sampled pool rather than a global optimum. The fixed baseline selects the single training-pool candidate with the lowest mean training CER, $\theta_{\mathrm{fixed},e}^{*} = \arg\min_{\theta_j \in \Theta} \frac{1}{|\mathcal{I}_{\mathrm{train}}|} \sum_{i \in \mathcal{I}_{\mathrm{train}}} y_{i,j,e}$. Random K samples K candidates uniformly from the same Θ, while NoPre uses the exact OCR run with all preprocessing operations disabled. We also define FixedTopK as a stronger static shortlist baseline: for each outer training fold, the K preprocessing candidates with the lowest mean training CER are selected and then applied as the same K-candidate shortlist to every held-out page.

4.2 Contour Suppression as Upstream Page-Level Preprocessing

As an optional preprocessing step, we use a fixed contour-based rule to identify rectangular regions that meet fixed criteria and replace their pixels with white before text detection. It is intended to reduce large non-text regions that can trigger

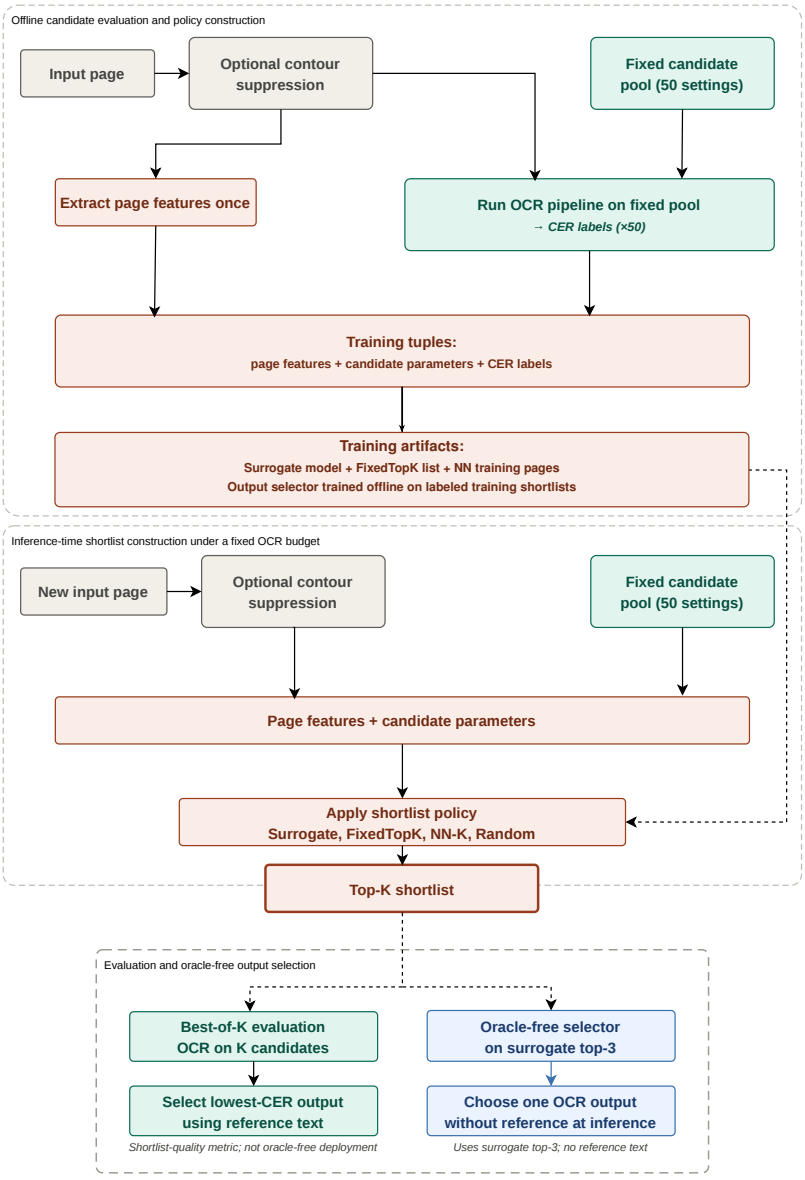


Fig. 1 Page-level preprocessing framework. Offline, each page is optionally contour-suppressed, page features are extracted once, and a fixed pool of 50 candidates is evaluated with OCR to obtain CER targets for recognizer-specific surrogate training and baseline construction. The output selector is trained offline on labeled training shortlists. At inference, shortlist policies rank or transfer candidates for a new page under a limited OCR budget. The lower branches distinguish reference-based best-of-K evaluation from selection of one surrogate top-3 output without reference text at inference

false detections, such as stamps, decorative headers, logos, and watermarks, rather than to serve as a general artifact-removal method. Given an input page image, the method converts the image to grayscale, applies a Gaussian blur with a 5-by-5 kernel, computes an Otsu threshold, and extracts external contours from the inverted threshold mask. For each contour, a bounding rectangle is computed; if the rectangle satisfies all of the following criteria, it is overwritten with white pixels in the original image: contour area greater than 500 pixels, aspect ratio between 1.0 and 5.0, width greater than 80 pixels, and height greater than 30 pixels. A pre-NFC exploratory check on 30 pages

with VietOCR found maximum absolute changes in mean CER of 0.06 pp when varying the upper aspect-ratio bound and 0.25–0.56 pp for the tested area and size thresholds (Online Resource 1); it was not repeated on final references or other recognizers. We therefore treat contour suppression as a fixed condition, not a tuned artifact-removal module. The output remains a page image rather than a binarized image; only selected large regions are suppressed. The objective is to reduce false detections before OCR, not to directly improve recognition in already detected text crops.

4.3 Page-Level Preprocessing Search Space

The page-level preprocessing search space has five tunable parameters applied to the grayscale page before text detection: `gaussian_ksize` $\in \{0, 3, 5, 7\}$, `bilateral_d` $\in \{0, \ldots, 9\}$, `clahe_clip` $\in [0, 4.0]$, `gamma` $\in [0.5, 1.5]$, and `sharpen` $\in [0, 2.0]$. These parameters correspond to Gaussian blur, bilateral filtering, CLAHE, gamma correction, and sharpening, respectively; their off/identity values and roles are tabulated in Online Resource 1.

These five controls were selected to address common page-level degradations using lightweight operations while keeping the search space compact and compatible with stratified sampling. Gaussian blur and bilateral filtering address noise under different smoothing assumptions, CLAHE targets local contrast and faded text, gamma correction adjusts global brightness, and sharpening addresses blurred strokes. We restrict the search to grayscale intensity operations that preserve page geometry and do not convert the image to a binary representation. Operations such as binarization and morphology would require categorical or pipeline-level design choices outside the fixed five-dimensional search space. Geometric corrections such as deskewing are outside this study's intensity-based preprocessing scope. A post hoc projection-profile skew diagnostic is reported in Online Resource 1.

The two denoising parameters are discrete: `gaussian_ksize=0` disables Gaussian blur, while positive values specify the Gaussian kernel size. Positive Gaussian kernel sizes are restricted to odd values, as required by the OpenCV implementation, and a kernel size of 1 is excluded because it is effectively an identity operation and redundant with the off value. Similarly, `bilateral_d=0` disables bilateral filtering, while positive values specify the bilateral filter diameter. A post hoc OpenCV 4.10 check found that several positive diameters produced identical outputs under the fixed sigmas; thus the ten encodings are not necessarily ten distinct transforms (Online Resource 1). The original candidate pool was retained. The remaining parameters are continuous, with `clahe_clip=0` disabling CLAHE, `gamma=1.0` serving as the identity brightness transform, and `sharpen=0` disabling sharpening.

These ranges were retained after boundary checks on 20 pages per check using an earlier reference version. These checks varied one parameter at a time across the three recognizers, covering both raw and contour-suppressed settings. The checks clearly supported the Gaussian upper bound; the other bounds were retained as common operating ranges rather than recognizer-specific optima.

Implementation settings outside this five-dimensional space are held fixed across all experiments: OpenCV automatic sigma for Gaussian blur, bilateral color and spatial sigma values of 75, an 8-by-8 CLAHE tile grid, and parametric 3-by-3 sharpening. Operations are applied in the fixed order Gaussian blur, bilateral filtering, CLAHE, gamma correction, and sharpening, with denoising before contrast/brightness adjustment and sharpening last to avoid amplifying earlier-stage noise.

4.4 Data Collection via Stratified Candidate Sampling

A fixed candidate pool of 50 preprocessing combinations is generated from a stratified five-dimensional unit-cube design, following the space-filling motivation of Latin hypercube sampling [12], and then reused for every page. The two discrete parameters are obtained by binning the corresponding design coordinates into the discrete values actually used: `gaussian_ksize` into {0, 3, 5, 7} and `bilateral_d` into {0, ..., 9}. The three continuous parameters (`clahe_clip`, `gamma`, and `sharpen`) are scaled linearly from their design coordinates into their stated ranges. This Latin-hypercube-style construction gives stratified one-dimensional coverage for the continuous parameters and near-balanced marginal counts for

the discrete levels, while treating the 50 candidates as a compact finite design rather than a full factorial grid. For each page and candidate combination, the page is preprocessed, OCR is run, and page-level CER is computed against the reference transcription. This yields $300 \times 50 = 15{,}000$ rows per recognizer in the primary dataset, one for each page i and candidate θ_j, with each row storing page features, preprocessing parameters, and the observed CER after OCR.

The candidate-pool design aims for broad coverage of the preprocessing response surface rather than per-image convergence to an optimum; with $D = 5$ search dimensions, $M = 50$ follows the common $n \approx 10d$ rule of thumb for initial space-filling computer experiments [11]. The sampled-pool oracle reported in the experiments therefore refers to the best observed configuration within this finite pool, not a global optimum over the continuous preprocessing space. The offline OCR evaluation loop is summarized in Online Resource 1.

4.5 Page-Level Features

For each page, OCR-free features are extracted before candidate evaluation, from the original page in the raw setting and the contour-suppressed page otherwise. Seven intensity and texture columns describe blur, contrast, brightness, Laplacian standard deviation (legacy field `noise`), edge density, entropy, and dynamic range. Blur is Laplacian variance; `noise` is its square root, not an independent noise estimate. Five further columns describe text density, skew angle, connected-component count, aspect ratio, and estimated line count. The 12-column representation is retained, including this deterministic redundancy.

The surrogate input concatenates these 12 page-level columns with five preprocessing parameters and five predefined interaction terms: blur × `gaussian_ksize`, contrast × `clahe_clip`, brightness × `gamma`, `noise` × `bilateral_d`, and entropy × `sharpen`. These fixed interactions combine page properties and controls without data-driven feature selection. The two Laplacian interactions use different controls and are not identical columns. The resulting input has 22 dimensions, not 22 independent quantities.

We use OCR-free hand-crafted page features rather than learned visual embeddings to keep the ranking model lightweight and independent of any additional pretrained image model at inference time. The feature definitions also allow domain-motivated interaction terms that directly pair image properties with the preprocessing controls they are intended to modulate; the full feature-group table is provided in Online Resource 1.

4.6 Surrogate Model

For each recognizer, we train a separate surrogate regressor to predict page-level CER for a given page and preprocessing candidate. The input is the 22-dimensional vector described above, and the target is the actual CER obtained after applying the candidate preprocessing setting and running the OCR pipeline. The surrogate is therefore recognizer-specific but shares the same feature representation and candidate pool across recognizers.

The surrogate is used for ranking rather than for exact CER prediction. For each held-out page, the model scores every candidate preprocessing combination in the fixed candidate pool, and candidates with lower predicted CER are ranked higher. We refer to this as a cross-instance surrogate because the model is trained across many pages and reused for unseen pages, rather than being fitted separately for each page through iterative feedback.

We use regression followed by ranking rather than a direct pairwise or listwise learning-to-rank objective for two practical reasons. First, the observed target is a continuous page-level CER, and continuous CER predictions are useful beyond candidate ordering: they define Top1Pred, prediction margins, and features for the oracle-free top-3 selector. Second, the main methodological question is whether page-conditioned scoring can enrich a small shortlist under a fixed OCR-call budget, not which tabular ranking loss is optimal. Direct learning-to-rank models are therefore left as an extension rather than treated as a primary contribution.

We consider four regression model families: Ridge regression, Random Forest, XGBoost, and LightGBM. These families cover linear, bagged-tree, and gradient-boosted regressors commonly

used for feature-based tabular regression. Model-family selection and evaluation are performed within the nested GroupKFold protocol described in Section 5. We do not include stacking in the primary evaluation because group-aware out-of-fold predictions would be needed to train a stacking meta-learner without mixing candidate rows from the same page.

4.7 Evaluation Protocol

We evaluate shortlist quality under fixed OCR-call budgets $K \in \{3, 5, 10, 15\}$. These values span tight to moderate budgets, covering 6% to 30% of the 50-candidate pool. For each held-out page, the surrogate ranks the fixed 50-candidate pool by predicted CER, and the Surrogate K shortlist consists of the K candidates with lowest predicted CER. Random K is the exact expected best-of-K CER under uniform sampling of K distinct candidates without replacement from the same pool; it is computed analytically from the ordered observed CER values, eliminating Monte Carlo noise. FixedTopK selects the K candidates with the lowest mean training CER in each outer fold and uses that same static shortlist for every held-out page. Surrogate K, Random K, and FixedTopK are evaluated using best-of-K CER, defined as the minimum actual CER among the K candidates in the shortlist.

As an additional retrieval-based baseline, we evaluate a nearest-neighbor transfer shortlist (NN-K) in the Vietnamese raw, contour-suppressed, and SROIE settings. For each outer fold, page features are standardized using only the training pages. For each held-out page, NN-K finds the nearest training pages in this OCR-free feature space, transfers the oracle-best preprocessing candidate from each nearest training page, removes duplicate candidates while preserving neighbor order, and fills any remaining shortlist slots with fold-specific FixedTopK candidates if needed. NN-K therefore uses only training-fold OCR responses for candidate transfer and is evaluated on held-out pages with the same best-of-K protocol as Surrogate K and FixedTopK.

We also report three reference baselines. NoPre bypasses grayscale conversion and the five controls, passing the color page to the OCR pipeline (after contour suppression in that setting). Pool candidates are converted to grayscale; the pool contains neither an identity nor a grayscale-only candidate. Gains over NoPre therefore include grayscale conversion and do not isolate the five controls. FixedBest selects the combination with lowest mean training CER in each fold. Sampled Oracle is the minimum observed CER among the 50 candidates, not a global parameter optimum.

We report computational cost primarily in OCR-call units because OCR runtime depends on hardware, recognizer implementation, and page content. The offline data collection stage requires 50 OCR evaluations per page and is used only to build the training and evaluation table. At inference time, OCR-free feature extraction and scoring the 50 tabular candidate rows require no additional OCR calls, but their wall-clock overhead is not benchmarked here. NoPre, FixedBest, and Top1Pred require one OCR call per page; the oracle-free top-3 selector requires three OCR calls; Surrogate K, NN-K, and FixedTopK require K OCR calls under the best-of-K shortlist-quality protocol; and the sampled-pool oracle requires all 50 OCR calls. OCR-call count therefore captures the repeated inference budget, while hardware-specific end-to-end latency remains outside the scope of this study.

Because Surrogate K and Random K both use best-of-K selection during evaluation, these results measure shortlist quality under a fixed candidate-evaluation budget rather than oracle-free output-selection performance. The comparison remains fair because both methods receive the same within-shortlist best-of-K evaluation. We additionally compute an oracle-free one-call operating point, Top1Pred, defined as the single candidate with the lowest out-of-fold predicted CER. Top1Pred does not use reference text or within-shortlist CER selection; it is reported as a supplementary single-output baseline rather than as the main shortlist-quality metric.

To test whether a short surrogate shortlist can be used without reference text at inference time, we also evaluate an oracle-free top-3 output selector. For each held-out page, OCR is run on the surrogate-ranked top-3 candidates and the selector chooses one of the three OCR outputs. Candidate-level selector features include the surrogate rank, predicted CER and prediction margins, preprocessing parameters, the number of detected crops, OCR text length and character-composition statistics, simple document-pattern

counts such as dates and administrative keywords, and agreement features based on normalized edit distances among the three OCR outputs. We compare five selector families: random-forest and logistic-regression classifiers trained to identify the best candidate within the top-3, and random-forest, gradient-boosting, and ridge regressors trained to predict candidate CER. Selector-family choice is nested inside each outer fold: the family with the lowest inner-fold validation CER on the outer-training pages is refit on all outer-training pages and then used on the held-out pages. The held-out selector therefore uses OCR outputs, surrogate predictions, and preprocessing metadata, but not reference text or held-out CER values. We compare it with Top1Pred, the agreement-only baseline, FixedBest, NoPre, and the best-of-3 oracle. The agreement-only baseline selects the top-3 output with the smallest mean normalized edit distance to the other two outputs.

For a feature-matched sensitivity analysis, we replace the surrogate's predicted CER and prediction-margin features by the within-shortlist ranks 1, 2, and 3. The exact surrogate top-3 candidates, OCR outputs, and all remaining selector features are retained, and the nested selector is rerun. This makes the prediction-related inputs comparable with NN-K and FixedTopK, which provide an ordering but not a continuous CER prediction. The analysis tests whether policy comparisons persist without the surrogate's numerical score and margin information. It is not a pure test of shortlist quality because the selector is refit and continues to use OCR-output features.

5 Experimental Setup

5.1 OCR Recognizers

All Vietnamese experiments use PaddleOCR with Vietnamese configuration (`lang=vi`) for initial text detection. The detected regions are cropped and passed to VietOCR, Tesseract with the Vietnamese language pack (`vie`), or EasyOCR with Vietnamese recognition (`vi`). The initial detector settings and preprocessing operations are shared, but crop-level processing differs: VietOCR recognizes the crop directly, EasyOCR `readtext` performs detection followed by recognition, and Tesseract uses its default page segmentation.

The final candidate-response data were generated with PaddleOCR 3.3.2 on PaddlePaddle GPU 3.3.0, VietOCR 0.3.13 using its `vgg_transformer` configuration, EasyOCR 1.7.2, and Tesseract 5.3.4 through `pytesseract` 0.3.13. The installed Tesseract `eng` and `vie` language packages were version `1:4.1.0-2`. For each page, detector polygons were converted to axis-aligned bounding-rectangle crops and processed in the `dt_polys` order returned by PaddleOCR; no additional reading-order sorting was applied. Nonempty recognized strings were joined with tab separators in that preserved order. VietOCR output was retained only when its predicted confidence was strictly greater than 0.8. Tesseract was called with its default engine and page-segmentation settings.

PaddleOCR was used for initial detection because it supports a modular crop-based pipeline. The backends were chosen to cover complementary practical OCR systems: VietOCR as a Vietnamese-specific neural recognizer, Tesseract as a mature open-source OCR engine with LSTM recognition, and EasyOCR as a general-purpose neural OCR toolkit. These backends test whether preprocessing-selection behavior changes across OCR systems, rather than benchmarking their best attainable accuracy. For SROIE, the same initial-detection and crop-processing structure is used with English configurations, and VietOCR is omitted because it is Vietnamese-specific.

5.2 Nested GroupKFold Cross-Validation

For the Vietnamese experiments, all reported surrogate results are computed with a nested 5-fold GroupKFold protocol, using the page image as the grouping unit. All 50 candidate preprocessing rows from the same page are kept in the same fold, preventing leakage in which candidates from one page appear in both training and test data. Each outer fold therefore evaluates the selected surrogate on held-out pages that were not used to fit that fold's model. After the five outer folds are completed, the held-out page-level results are combined across folds, giving one out-of-fold result for each of the 300 pages. The same group-aware nested protocol is used for the SROIE setting, with results combined over all 612 receipt images.

Model-family selection is nested inside each outer training fold. For each recognizer and outer fold, the candidate model families are compared using inner GroupKFold on the outer-training pages, and the family with the lowest mean validation MAE is selected. That selected family is then refit on the full outer-training fold and evaluated on the corresponding held-out fold. This nested protocol avoids selecting a model family using information from the outer test pages, following standard guidance on reducing selection-induced optimism in cross-validation [3, 16, 29].

A final model is also selected and refit on all available pages, but this final model is not used to compute the reported cross-validation metrics. A compact protocol table and stepwise nested-evaluation algorithm are provided in Online Resource 1.

5.3 Metrics

CER is computed after symmetric Unicode NFC and whitespace normalization. Vietnamese evaluation retains letter case and uses the frozen corrected reference version v3. SROIE additionally applies Unicode casefold to both predictions and references; accents and punctuation are retained. Surrogate prediction quality is summarized by MAE and R^2. Global Spearman is computed over held-out candidate rows within each outer fold; per-image Spearman is the mean within-page candidate-rank correlation. Best-of-K reports the lowest observed CER in each selected shortlist, not a deployable output choice.

For statistical comparison, Surrogate K is compared with Random K and FixedTopK using two-sided paired Wilcoxon signed-rank tests over page-level best-of-K CER values. The supplementary NN-K baseline is compared with Surrogate K, FixedTopK, and Random K using the same two-sided paired page-level tests. The oracle-free top-3 selector is compared with Top1Pred, the agreement-only baseline, FixedBest, and NoPre using two-sided paired page-level tests. Complete policy-matched one-output comparisons additionally contrast the Surrogate selector with selectors fitted to NN-K and FixedTopK shortlists. Tests are run separately for each recognizer, setting, and K value where applicable, and the direction of an effect is reported from the paired mean CER difference. Aggregate CER values are reported as means over the held-out page-level results combined across outer folds. We also compute percentile bootstrap confidence intervals for paired mean differences, signed as the first named method minus its comparator. Random K is the exact expected minimum CER under uniform sampling of K distinct candidates without replacement from the 50-candidate pool, computed from the ordered observed CER values rather than Monte Carlo draws.

The Wilcoxon test uses SciPy's `zero_method=wilcox` convention, which excludes zero paired differences from the signed ranks. This convention matters when shortlisted policies tie on many pages; a zero-handling sensitivity is reported below. The test assesses paired signed-rank evidence, not a null hypothesis about the mean difference; the mean difference and its bootstrap interval describe effect size. These 95% intervals are pointwise and are not Holm-adjusted, so an interval excluding zero does not by itself establish significance after multiplicity correction.

For the retrospective headroom analysis, let $F_{i,K}$ be the fold-specific FixedTopK CER, O_i the sampled-pool-oracle CER, and $P_{i,K}$ the CER of policy P. We define headroom $H_{i,K} = F_{i,K} - O_i$ and FixedTopK-relative policy advantage $A^P_{i,K} = F_{i,K} - P_{i,K}$. These quantities share $F_{i,K}$, and $A^P_{i,K} \leq H_{i,K}$ because the sampled-pool oracle is the best candidate in the pool. We therefore compute the coupled correlation for Surrogate, Random, and NN-K, and separately correlate headroom with direct policy-to-policy gains that do not contain FixedTopK: Random minus Surrogate, Random minus NN-K, and NN-K minus Surrogate. This control analysis is descriptive and is not used for model or policy selection.

Because the study reports multiple paired comparisons, we control multiplicity with the Holm procedure within explicitly defined scientific comparison families. For best-of-K analyses, one family is one policy contrast within one evaluation block: 12 recognizer-budget tests for raw Vietnamese, 12 for contour-suppressed Vietnamese, or eight for SROIE. For the oracle-free selector versus Top1Pred, agreement-only, FixedBest, and NoPre, each comparator defines one eight-test family across the setting–recognizer combinations. The primary policy-matched Surrogate-versus-NN-K/FixedTopK comparison forms one 16-test family; the rank-only sensitivity forms a

separate 16-test family. For the headroom analysis, the 32 correlations for each coupled policy outcome or direct non-shared outcome define a family. The public statistical tables report both unadjusted and Holm-adjusted p-values; claims of statistical significance use the Holm-adjusted values unless explicitly labeled unadjusted. We also emphasize effect sizes and bootstrap confidence intervals and avoid drawing strong conclusions from isolated marginal tests.

5.4 Hyperparameters and Sensitivity

The primary results use the nested GroupKFold protocol described above, with model-family selection performed inside each outer training fold. For implementation-level regressor settings, we do not run a broad hyperparameter search. Instead, we compare four predefined surrogate model families, chosen to cover common linear, bagged-tree, and gradient-boosted regressors, with fixed hyperparameters for each family. The goal is not to exhaustively benchmark surrogate regressors, but to evaluate whether a practical cross-instance surrogate can provide useful shortlist rankings. This keeps the focus on candidate ranking under a fixed OCR budget rather than on maximizing a standalone regressor benchmark.

5.5 Use of AI Tools

Large language models—Claude (Anthropic) and Codex (OpenAI)—assisted with drafting and revising analysis and evaluation scripts and some manuscript text. They also helped flag possible reference-transcription anomalies using character scans and OCR disagreements, prepare candidate corrections, and inspect selected source-image regions. AI- or OCR-suggested text was not treated as ground truth by itself: the authors approved adopted label edits, with targeted image checks documented in the label-remediation record. This process did not constitute exhaustive proofreading of every transcription. The authors take responsibility for the final references, analyses, and manuscript.

6 Results

6.1 Preprocessing Response Variation in the Fixed Candidate Pool

Initial PaddleOCR detection settings are held fixed within each dataset, while the backend-specific crop processing described above is retained. We next quantify the observed preprocessing-response variation within the fixed candidate pool.

The 50-candidate pool exhibits substantial within-page variation. Mean within-image CER coefficients of variation are 57.2%, 68.8%, and 25.5% for raw VietOCR, Tesseract, and EasyOCR, and 71.1%, 79.4%, and 29.7% after contour suppression. The sampled-pool oracle improves over exact NoPre on 90–98% of Vietnamese pages, with mean per-page relative CER reductions of 15.4–39.4% across the six settings (full results in Online Resource 1). This oracle evaluates all 50 candidates and is not deployable; it establishes that the pool contains useful choices that a budgeted policy should retrieve.

6.2 Surrogate Model Quality

The surrogate models show moderate CER prediction accuracy and rank correlation. Across the six Vietnamese recognizer-setting combinations, nested-CV R^2 ranges from 0.467 to 0.589, global Spearman from 0.576 to 0.678, and mean per-image Spearman from 0.510 to 0.781; full model-quality aggregates are given in Online Resource 1. These are saved outer-fold aggregates, not a new independent reconstruction of out-of-fold predictions. Per-image Spearman is most directly aligned with ranking the 50 candidates separately for each page. We next evaluate shortlist quality directly using best-of-K CER.

6.3 Surrogate Shortlists vs Random Shortlists

Under the same K-candidate budget, two-sided paired Wilcoxon tests show lower CER for surrogate shortlists than for random shortlists across all three recognizers, all evaluated K values, and both raw and contour-suppressed settings. All 12/12 raw and 12/12 contour-suppressed surrogate-versus-random comparisons remain significant

after Holm correction within their respective evaluation-block families, giving 24/24 adjusted-significant comparisons overall. Best-of-K curves across recognizers and budgets are visualized in Online Resource 1.

The effect-size pattern is consistent with the budgeted-ranking interpretation: surrogate gains over random are largest at small K and generally decrease as K grows, because random shortlists become more likely to contain a good candidate when the shortlist budget increases. The gains are largest for EasyOCR and smallest for VietOCR, which has the lowest NoPre CER in the Vietnamese experiments. The stronger FixedTopK baseline is more competitive than Random K, however, and is analyzed separately below.

6.4 Comparison with FixedTopK Static Shortlists

Table 2 compares the surrogate with FixedTopK. In raw Vietnamese, one of 12 unadjusted two-sided tests favors the surrogate, but none remains significant after Holm correction. In contour Vietnamese, one unadjusted test favors the surrogate and two favor FixedTopK; after Holm correction, EasyOCR $K = 15$ favors the surrogate (-0.07 pp) and VietOCR $K = 3$ favors FixedTopK ($+0.09$ pp). On SROIE, all eight surrogate improvements remain significant. Pointwise bootstrap intervals for Surrogate minus FixedTopK favor the surrogate in 1/12 raw, 1/12 contour and 8/8 SROIE comparisons, and favor FixedTopK in 1/12 raw and 2/12 contour comparisons. Illustrative differences are raw Tesseract $K = 3$: $+0.20$ pp [0.04, 0.37]; contour VietOCR $K = 3$: $+0.09$ pp [0.01, 0.17]; and SROIE Tesseract $K = 3$: -1.36 pp [-1.69, -1.03]. These 95% intervals are pointwise, not Holm-adjusted. EasyOCR $K = 15$ has 215/300 ties in raw and 216/300 in contour; its significance conclusion depends on Wilcoxon zero handling in both settings (Online Resource 1). The adverse contour VietOCR $K = 3$ result remains significant under all three checked zero conventions with the primary features (primary convention: `wilcox`). However, neither this disadvantage nor the contour EasyOCR $K = 15$ advantage remained significant after removing the redundant column (Online Resource 1).

6.5 Policy-Agnostic FixedTopK-to-Oracle Headroom

The aggregate comparison above shows that FixedTopK leaves different amounts of residual improvement across pages and settings. We quantify this as the per-page headroom $H_{i,K} = F_{i,K} - O_i$. A surrogate-only analysis would compare it with $F_{i,K} - S_{i,K}$, the surrogate's advantage over FixedTopK. However, the two quantities share $F_{i,K}$, and the sampled-pool oracle enforces $F_{i,K} - P_{i,K} \leq H_{i,K}$ for every pool-constrained policy P. A positive correlation can therefore reflect this construction rather than a learned-ranking effect.

We assess the relationship using the same held-out pages for Surrogate, Random, and NN-K. As Table 3 shows, each coupled FixedTopK-relative advantage has a significantly positive correlation with headroom in all 32 combinations of recognizer, setting, and budget after Holm correction within each 32-correlation policy family, and the highest-headroom quartile has a larger mean advantage than the lowest quartile in all 32 configurations for every policy. Random's correlation is larger than the surrogate's in 30/32 configurations. Thus, the original 32/32 pattern is policy-agnostic and cannot establish that high-headroom pages specifically favor learned ranking.

We additionally correlate headroom with direct policy-to-policy gains that do not contain FixedTopK. Headroom versus Random minus Surrogate ranges from -0.502 to -0.057: it is negative in all 32 configurations and significantly negative in 28/32 after Holm correction. Headroom versus NN-K minus Surrogate is also negative in all 32 configurations. The Random-minus-NN-K control is mixed: two correlations are significantly positive before correction and none remains positive after Holm correction across its 32-correlation family. These controls do not support using headroom alone to select among policies.

Headroom remains a valid retrospective description of how far FixedTopK lies from the best candidate observed in the finite pool. It requires labeled historical data and evaluation of the pool, and it does not identify which alternative policy will achieve that improvement. Evidence about policy quality therefore comes from the direct mean-CER, paired-test, and oracle-free selector comparisons reported in the surrounding sections.

Table 1 Vietnamese best-of-K CER results. Values are percentages. Each S/R entry reports Surrogate K / Random K.

Setting	Recognizer	NoPre	FixedBest	SampOracle	K=3 S/R	K=5 S/R	K=10 S/R	K=15 S/R
Raw	VietOCR	4.59	4.57	3.50	4.18 / 4.48	4.00 / 4.16	3.79 / 3.88	3.71 / 3.75
Raw	Tesseract	9.49	9.03	6.51	7.93 / 8.64	7.50 / 8.02	7.06 / 7.41	6.88 / 7.13
Raw	EasyOCR	15.24	14.99	13.01	14.10 / 15.79	13.75 / 14.97	13.42 / 14.19	13.23 / 13.84
Contour	VietOCR	3.06	3.12	2.21	2.85 / 3.04	2.66 / 2.75	2.48 / 2.51	2.39 / 2.41
Contour	Tesseract	7.39	6.92	4.88	6.00 / 6.68	5.65 / 6.11	5.25 / 5.58	5.12 / 5.35
Contour	EasyOCR	13.07	12.90	10.93	12.13 / 13.72	11.73 / 12.88	11.32 / 12.10	11.13 / 11.75

Table 2 Comparison with FixedTopK static shortlists. CER values are percentages. Diff is Surrogate minus FixedTopK, so negative values favor the surrogate.

Setting	Recognizer	Surr K=3	FixedTopK K=3	Diff K=3	Surr K=15	FixedTopK K=15	Diff K=15
Raw	VietOCR	4.18	4.16	+0.03	3.71	3.73	-0.03
Raw	Tesseract	7.93	7.72	+0.20	6.88	6.92	-0.04
Raw	EasyOCR	14.10	14.13	-0.03	13.23	13.32	-0.09
Contour	VietOCR	2.85	2.76	+0.09	2.39	2.42	-0.03
Contour	Tesseract	6.00	5.83	+0.16	5.12	5.14	-0.03
Contour	EasyOCR	12.13	12.06	+0.07	11.13	11.20	-0.07
SROIE	Tesseract	24.07	25.43	-1.36	21.70	22.25	-0.54
SROIE	EasyOCR	18.86	19.22	-0.36	18.02	18.21	-0.19

6.6 Nearest-Neighbor Transfer Baseline

NN-K outperforms Random K in all evaluated comparisons after Holm correction within each block. Neither page-conditioned policy dominates. No raw Vietnamese NN-K–Surrogate comparison survives Holm correction (smallest adjusted $p = 0.0818$). On contour VietOCR, NN-K is significantly better than Surrogate at $K = 3, 5, 10$ by 0.13, 0.12 and 0.10 pp, respectively. On SROIE Tesseract, Surrogate is significantly better at $K = 3, 5, 10$; on SROIE EasyOCR, NN-K is significantly better at $K = 3, 5$. Holm-adjusted counts are summarized in Online Resource 1; full paired comparisons are in the public tables. After removing the redundant column, the contour VietOCR $K = 10$ difference lost significance and the SROIE Tesseract $K = 15$ Surrogate advantage became significant (post hoc analysis in Online Resource 1).

The SROIE comparisons above use casefolded CER. A matched case-sensitive analysis on the corrected 612-receipt cohort is retained in Online Resource 1: NN-K's EasyOCR $K = 3$ advantage is not significant under that metric ($p_H = 0.0778$, versus 0.0336 after casefolding). Threshold crossings do not establish a metric-by-policy interaction.

6.7 VietOCR in the Lowest-Baseline-CER Setting

Despite its low baseline CER, VietOCR's oracle-assisted best-of-K shortlist improves at $K = 3$: raw CER is 4.18% for Surrogate versus 4.59% NoPre, 4.57% FixedBest and 4.48% Random; contour CER is 2.85% versus 3.06%, 3.12% and 3.04%, respectively. Best-of-K requires reference CER to choose within the shortlist and is not deployable one-output CER.

6.8 Contour Suppression as a Complementary Stage

Contour suppression removes large rectangular regions before detection, whereas the surrogate ranks photometric settings. One-call NoPre CER decreases from 4.59% to 3.06% (VietOCR), 9.49% to 7.39% (Tesseract), and 15.24% to 13.07% (EasyOCR). The separate oracle-assisted contour + Surrogate $K = 15$ evaluation reaches 2.39%, 5.12% and 11.13%. These are different output-selection protocols; the latter values are not deployable one-output gains.

Table 3 Policy-agnostic headroom controls over 32 dataset/setting/recognizer/budget configurations. Coupled outcomes share FixedTopK with headroom; non-shared outcomes are direct policy-to-policy gains. Positive sig. counts significantly positive Spearman correlations after Holm correction within each 32-correlation outcome family; Q4>Q1 counts configurations whose highest-headroom quartile has a larger mean outcome than the lowest.

Analysis	Outcome	Spearman r range	Positive sig.	Q4>Q1
Coupled	FixedTopK − Surrogate	[0.336, 0.686]	32/32	32/32
Coupled	FixedTopK − Random	[0.384, 0.844]	32/32	32/32
Coupled	FixedTopK − NN-K	[0.376, 0.764]	32/32	32/32
Non-shared	Random − Surrogate	[-0.502, -0.057]	0/32	2/32
Non-shared	Random − NN-K	[-0.265, 0.155]	0/32	13/32
Non-shared	NN-K − Surrogate	[-0.404, -0.077]	0/32	0/32

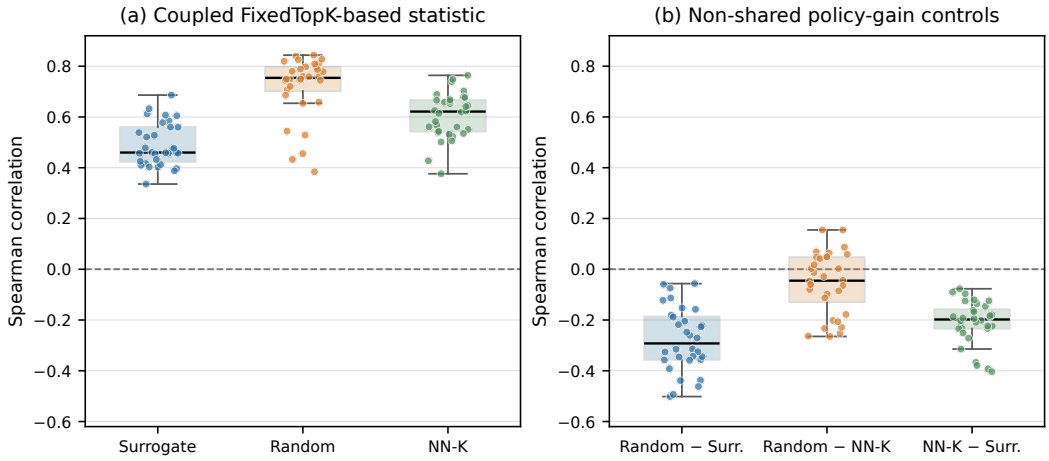


Fig. 2 Spearman correlations across all 32 dataset/setting/recognizer/budget configurations. Each point is one configuration and the dashed line marks zero. (a) Headroom versus FixedTopK-relative advantage is positive for Surrogate, Random, and NN-K, consistent with the shared FixedTopK term and oracle constraint. (b) Direct policy-to-policy gains remove the shared term and show no corresponding positive pattern for learned ranking. Boxes show the median and interquartile range

6.9 SROIE Evaluation: Surrogate Outperforms FixedTopK

The SROIE evaluation applies the same full-page shortlist-ranking protocol to a different document domain and English OCR configuration. Unlike the Vietnamese settings, the surrogate directly outperforms FixedTopK across all evaluated SROIE recognizer-budget comparisons. SROIE also has greater aggregate FixedTopK-to-oracle headroom, but Section 6.5 shows that headroom alone does not explain which policy will perform best.

The omitted $K = 5, 10$ rows follow the same pattern and appear in Online Resource 1. All eight surrogate-versus-random and surrogate-versus-FixedTopK tests favor the surrogate and remain significant after Holm correction. All eight NN-K comparisons with Random and FixedTopK also remain significant, but its ordering against the surrogate depends on recognizer and budget. These full-page, casefolded-CER results are not receipt key-information-extraction or cropped-field benchmarks.

6.10 Oracle-Free Top-3 Output Selection

Best-of-K uses observed within-shortlist CER. As an oracle-free check, Table 5 reports the nested selector using only the surrogate top-3 OCR outputs and derived prediction, text-quality, and agreement features. Its comparisons with

Table 4 SROIE receipt-setting results. CER values are percentages. Surrogate model-quality metrics are reported in Online Resource 1.

Recognizer	K	NoPre	FixedBest	SampOracle	Surrogate	NN-K	FixedTopK	Random
Tesseract	3	31.99	28.12	20.88	24.07	24.68	25.43	28.52
Tesseract	15	31.99	28.12	20.88	21.70	21.80	22.25	23.04
EasyOCR	3	21.23	19.99	17.76	18.86	18.71	19.22	21.02
EasyOCR	15	21.23	19.99	17.76	18.02	18.01	18.21	18.53

Top1Pred, agreement-only, FixedBest, and NoPre are one-output CERs.

The selector has lower mean CER than Top1Pred in all eight settings, with four differences significant after Holm correction. Against FixedBest, mean CER is lower in five settings, with three significant differences and no significant increases. The significant FixedBest reductions occur for raw VietOCR, raw Tesseract and SROIE Tesseract. The largest is SROIE Tesseract: 25.15% versus 28.12% FixedBest. Top-3 output selection recovers part, but not all, of the best-of-3 oracle gain.

6.11 Policy-Matched Selector Comparisons and Feature-Matched Sensitivity

To compare complete deployable policies, we also fit the nested output selector to NN-K and FixedTopK shortlists. In the full-feature 16-test Holm family, Surrogate is significantly better than NN-K on raw Tesseract (−0.50 pp) and SROIE Tesseract (−0.45 pp), and better than FixedTopK on SROIE Tesseract (−1.18 pp). It is significantly worse than FixedTopK on contour VietOCR (+0.22 pp). The other 12 rows are not significant. These one-output results do not inherit the eight-of-eight SROIE best-of-K advantage over FixedTopK.

Replacing continuous surrogate predictions and margins with shortlist ranks preserves the top-3 candidates while matching the order-only inputs of NN-K and FixedTopK. The refitted rank-only selector forms a separate 16-test Holm family. It is significantly better than NN-K on raw Tesseract (−0.49 pp) and SROIE Tesseract (−0.46 pp), and better than FixedTopK on SROIE Tesseract (−1.20 pp), with no significant disadvantages. Because the selector is refit and retains OCR-output features, this is not a pure shortlist-quality comparison (full results in Online Resource 1).

7 Discussion

7.1 Ranking Utility Despite Moderate R-squared

Moderate R^2 does not preclude useful shortlists because the objective is to place strong candidates near the top, not to estimate every CER exactly. The nested per-image rank correlations and consistently better-than-random shortlists show that within-page ordering can suffice. However, inner-loop model-family selection minimizes candidate-level MAE rather than a within-page ranking or best-of-K loss, so the selection criterion is only indirectly aligned with the downstream objective.

7.2 Interpreting Policy Comparisons and Available Headroom

FixedTopK transfers well on the single-portal Vietnamese pages, whereas the surrogate consistently improves over it on SROIE; NN-K is also competitive. This setting-dependent pattern precludes a universal policy ranking. Small absolute differences should not be equated with practical improvements solely because a paired test is significant.

FixedTopK-to-oracle headroom indicates residual improvement in the sampled pool, but its correlation with every FixedTopK-relative advantage, including Random, can arise from the shared term and oracle constraint. It therefore requires labeled full-pool data and cannot choose a policy; direct held-out comparisons remain necessary.

Table 5 Oracle-free top-3 output-selection results. CER values are percentages. Sel-Fixed is Selector minus FixedBest, so negative values favor the selector. Raw and Holm-adjusted p-values are two-sided paired Wilcoxon tests for Selector versus FixedBest; Holm correction is across the eight rows.

Setting	Recognizer	Selector	Top1Pred	Agreement	FixedBest	NoPre	Sel-Fixed	Raw p	Holm p
Raw	VietOCR	4.48	4.58	4.51	4.57	4.59	-0.09	0.00329	0.0198
Raw	Tesseract	8.57	9.04	8.70	9.03	9.49	-0.45	0.000143	0.00100
Raw	EasyOCR	14.99	15.15	15.13	14.99	15.24	+0.00	0.483	0.966
Contour	VietOCR	3.22	3.24	3.15	3.12	3.06	+0.10	0.524	0.966
Contour	Tesseract	6.68	7.07	6.66	6.92	7.39	-0.24	0.0184	0.0919
Contour	EasyOCR	12.97	13.04	13.10	12.90	13.07	+0.07	0.190	0.570
SROIE	Tesseract	25.15	27.26	26.75	28.12	31.99	-2.98	1.55×10^{-38}	1.24×10^{-37}
SROIE	EasyOCR	19.78	20.01	19.73	19.99	21.23	-0.21	0.0435	0.174

7.3 Surrogate Ranking and Retrieval-Based Transfer

NN-K transfers candidates from similar pages rather than learning a CER function over (i, θ_j). Its advantage over Random does not by itself establish the contribution of page-feature matching. Its mixed comparison with the surrogate suggests complementary empirical behavior rather than causal superiority of either method.

At the best-of-K level, Surrogate versus NN-K remains setting-dependent. At the output-selection level, the full-feature selector has three significant advantages and one disadvantage in the 16-test policy-matched family. The rank-only sensitivity has three significant advantages and no significant disadvantages, without establishing universal superiority. Both variants are refit and retain OCR-output features.

7.4 Practical Interpretation of K

K is a shortlist-quality budget: the four policies report the best observed CER within a K-candidate shortlist, not a deployable one-output choice. Top1Pred is a one-call operating point, whereas the nested selector produces one output from three OCR calls.

8 Limitations

1. Best-of-K measures shortlist quality, not oracle-free deployment. The supervised top-3 selector uses three OCR calls and OCR-output features; feature matching removes its numerical score advantage but is still not a pure shortlist test.
2. No non-random policy wins universally: FixedTopK is strong on Vietnamese data, and NN-K is competitive in several settings.
3. External validity is limited to 300 first pages from one Vietnamese portal and 612 English-configured SROIE receipts; the same-receipt screen does not guarantee exhaustive duplicate recall.
4. The method selects five grayscale preprocessing controls before a fixed PaddleOCR detector; color, binarization, morphology, geometry, and adaptive pools remain outside scope. The effect of deskewing on OCR was not evaluated.
5. Detected regions are concatenated in the order returned by PaddleOCR, without a separate reading-order model. Page-level CER therefore reflects both text recognition and detector-output ordering.
6. Ranking quality varies by recognizer and fold; Tesseract's lower per-image than global Spearman shows that cross-page variation contributes to the global statistic.
7. Inner-loop surrogate model-family selection minimizes candidate-level MAE, whereas the downstream endpoint is within-page ranking and best-of-K CER. Nesting prevents test-fold leakage but does not guarantee that the MAE-selected family is optimal for shortlist ranking.
8. Page-level out-of-fold estimates combine predictions from models whose training folds overlap. The paired tests and page-resampling intervals do not explicitly account for this shared-model dependence or possible similarity among pages from the same source or template; their uncertainty estimates should therefore not be read as guarantees for new sources or templates.

9. Headroom is retrospective, needs labeled full-pool evaluations, and cannot identify the best alternative policy.

9 Conclusion

This paper presented a budgeted framework for page-conditioned selection of OCR preprocessing candidates from a fixed pool. Surrogate and NN-K shortlists beat random selection in every evaluated comparison after Holm correction, but FixedTopK remained strong on Vietnamese pages and no non-random policy won universally; all eight SROIE surrogate-versus-FixedTopK best-of-K comparisons remained significant after Holm correction. The strict nested top-3 selector achieved lower mean CER than FixedBest in five of eight settings, with three differences significant after Holm correction across the eight comparisons. In the separate 16-test policy-matched one-output family, the full-feature Surrogate selector had three significant advantages and one significant disadvantage; the rank-only sensitivity likewise did not establish universal superiority. The headroom controls showed that FixedTopK-to-oracle headroom is policy-agnostic and retrospective. Future work should investigate direct learning-to-rank objectives, broader preprocessing pools, explicit reading-order reconstruction, and latency-aware oracle-free output selection across additional document domains.

Statements and Declarations

Funding. This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Competing Interests. The authors have no relevant financial or non-financial interests to disclose.

Data availability. The primary document images are derived from public materials downloaded from the Vietnamese Government document portal (https://chinhphu.vn/he-thong-van-ban). The full derived image set and full reference transcriptions are not publicly redistributed because redistribution permission has not been obtained. The public release contains candidate-pool definitions, OCR-free derived page features, candidate metadata, sanitized evaluation outputs, and statistical-test summaries sufficient to reproduce the reported aggregate analyses; the public repository is available at https://github.com/virustous2/budgeted-ocr-preprocessing-selection. The SROIE receipt dataset follows its own benchmark distribution terms and is not redistributed with this manuscript.

Code availability. The code used for preprocessing candidate generation, surrogate evaluation, baseline comparisons, statistical summaries, and the public derived result tables are available in a public repository under the Apache License, Version 2.0, at https://github.com/virustous2/budgeted-ocr-preprocessing-selection. Raw images, full reference transcriptions, OCR text caches, crop images, and trained model pickle files are excluded from the repository.

Ethics approval. Not applicable.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

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